

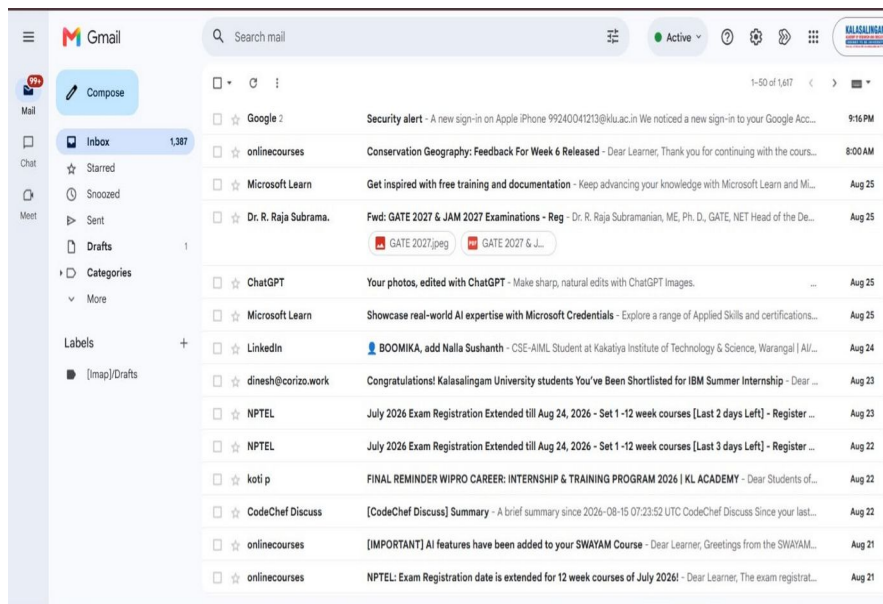
Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

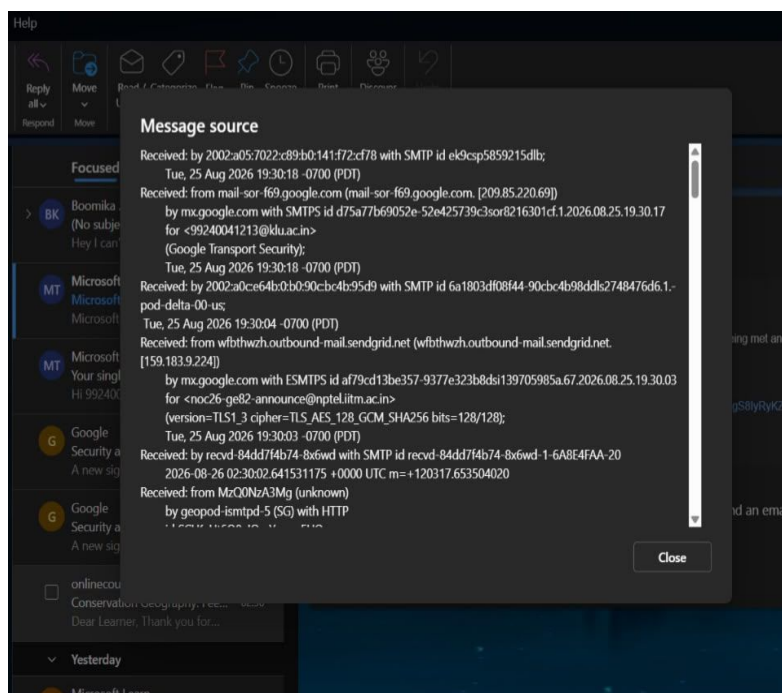
Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, "showoriginal"



- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers box"



- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

Original Message

Message ID	<010201a03e450416-aea8f0ac-303c-4ec8-9837-417ee631735b-000000@eu-west-1.amazonaws.com>
Created at:	Wed, Aug 26, 2026 at 7:01 PM (Delivered after 1 second)
From:	PortSwigger <noreply@portswigger.net>
To:	99240040319@klu.ac.in
Subject:	Your verification code
SPF:	PASS with IP 23.249.220.35 Learn more
DKIM:	'PASS' with domain portswigger.net Learn more
DMARC:	'PASS' Learn more

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.

KARE|SoC|CSE|2024-2025|Digital Forensics|Lab|Ex.No.4

Original Message

Message ID	<VB2JSn31pzn_NFVs8yKalA@notifications.google.com>
Created at:	Wed, Aug 26, 2026 at 9:16 PM (Delivered after 3 seconds)
From:	Google <no-reply@accounts.google.com>
To:	99240041213@klu.ac.in
Subject:	Security alert
SPF:	PASS with IP 209.85.220.73 Learn more
DKIM:	'PASS' with domain accounts.google.com Learn more
DMARC:	'PASS' Learn more

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- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

Message ID	<VB2JSn31pzn_NFVs8yKalA@notifications.google.com>
Created at:	Wed, Aug 26, 2026 at 9:16 PM (Delivered after 3 seconds)
From:	Google <no-reply@accounts.google.com>
To:	99240041213@klu.ac.in
Subject:	Security alert
SPF:	PASS with IP 209.85.220.73 Learn more
DKIM:	'PASS' with domain accounts.google.com Learn more
DMARC:	'PASS' Learn more

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.

```
Delivered-To: 99240041213@klu.ac.in
Received: by 2002:a05:7022:3d18:b0:141:f72:cf78 with SMTP id ui24csp13011dlb; Wed, 26 Aug 2026 08:46:49 -0700 (PDT)
X-Received: by 2002:a17:90b:2c88:b0:380:f85c:94b4 with SMTP id 98e67ed59e1d1-3966d20e7aamr18203644a91.7.1787759208712; Wed, 26 Aug 2026 08:46:48 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1787759208; cv=none; d=google.com; s=arc-20260327; b=Fz0kYeJgKQffBdpA6AzCyYACmZKubWbnIsjwsuVfQbwc7pXF44vB1d79qGzKexZrb/CXlpiTMIyKC0R9GZmc0AwJh1zW4350AYoA9t0ESzMTgVJvughh9DDIraCaEgRKJkEXZ2 0QKyORXX3KnjoOp7+n3gBrVUYrmW7JvVSqAwBDBZwcZe/31E9vh63o8EqT9f9ZpoLUS0u0XT7duFcuMnbteOylxmVGzuP9pjtcJ57bPBHQvnb84JxysvH11/V533Gdu198XC0wsa VnTV037LoAAK8b7Zoh3BrX/us9RIA1IR31Mu770ToaSubdlH1m4EHnH5FI4iqcSVsH0 x9nw==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=to:from:subject:message-id:globalsai:feedback-id:date:mime-
```



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 **Email Header Analyzer**

Paste Header:

```
; text-align: center; ><div> you received this email to let you know about i=
mportant changes to your Google Account and services.</div><div style=3D"di=
rection: ltr;">&copy; 2026 Google LLC, <a class=3D"afaf" style=3D"font-fami=
ly: &#39;Google Sans Flex&#39;, &#39;Google Sans Text&#39;, &#39;Google Sans&=
#39;, &#39;Noto Sans&#39;, Arial, Helvetica, sans-serif; color: rgba(0,0,0,0.54)=
; font-size: 11px; line-height: 18px; padding-top: 12px; text-align: center=
;">1600 Amphitheatre Parkway, Mountain View, CA 94043, USA</a></div><div><=
/div></td><td width=3D"8" style=3D"width: 8px;"></td></table></td></tr>
><tr height=3D"32" style=3D"height: 32px;"><td></td></tr></table></body></h=
tml>
--0000000000008c470e0659f52021--
```

Analyze Header

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

Message-ID	<VB2JSn31pzn_NFVs8yKaIA@notifications.google.com>
Subject	Security alert
From	Google <no-reply@accounts.google.com>
To	99240041213@klu.ac.in
Content-Type	multipart/alternative; boundary="00000000000008c470e0659f52021"



Domains Hosting Websites Email Security Whois Deals

209.85.220.73

Whois IP 209.85.220.73

Updated 1 second ago

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange: 209.85.128.0 - 209.85.255.255
CIDR: 209.85.128.0/17
NetName: GOOGLE
NetHandle: NET-209-85-128-0-1
Parent: NET209 (NET-209-0-0-0-0)
NetType: Direct Allocation
OriginAS:
Organization: Google LLC (GOGL)
RegDate: 2006-01-13
Updated: 2012-02-24
Ref: https://rdap.arin.net/registry/ip/209.85.128.0

OrgName: Google LLC
OrgId: GOGL
Address: 1600 Amphitheatre Parkway
```

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - o **Pass:** Indicates the IP is authorized.
 - o **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - o **Pass:** The email has not been tampered with.
 - o **Fail:** The email content may have been altered.

- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
- **Pass:** Indicates alignment with SPF/DKIM policies.
- **Fail:** Indicates possible spoofing or tampering.

DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=accounts.google.com; s=20251104; t=1787759208; x=1788364008; y=1788364008; dm=ku.ac.in; h=content-type; from:subject:message-id:gmsai-feedback-id:date:mime-version:from:to:cc:subject:date:message-id:reply-to; content-type; bh=anF4m0bUJLhzrIS/8q9KClp5BFNfHmSF80wCKDY=: b=GF9MzZaXU+ZYnENCpfr0xgnyVb8DNlkmwVFOTG5PeVxkZkQ65f0syuLkqyRm1 38dZuSa+BSBCYs+6Y1W7oqWG1B1wm5z960U+MeKIR+bSSeX8XAluL8EzLSWneL7XR w8wC1/hHBQ19lueUPBLWV+ma47XvUAPvAOzVtobc3kzY9DmBcvZm3Fic59dDHT.JnCjXDcoj/He9d7phgFkZzZ7hvcOP5vIvSVXl45a+qJpY6hu+SwbBar9A1QWoyzu ZkYyQjEXsWmXDDsc6Qhccqy1l6O1xFlVjPfuGU0FYs+fozbAupccDP2VnrOSqW6rQ HzUQ==
X-Google-DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=1e100.net; s=20251104; t=1787759208; x=1788364008; y=1788364008; h=content-type; from:subject:message-id:gmsai-feedback-id:date:mime-version:x-gm-message-state:from:to:cc:subject:date:message-id:reply-to; content-type; bh=anF4m0bUJLhzrIS/8q9KClp5BFNfHmSF80wCKDY=: b=cL2e6ymtePGfGmsMubOXClbtE2Aq5Lr49zlnof52eA4MTvTHNX1O5f5d60xz19r6 R04bskDNvd4IY98yH+keoycrJ1nsAjpqCv3wcQubhP5FNWKCWyzc39GuQAKJW+DU YMAInGnz6fpxD2wu5HNZqZ34waBdQl8CnMc5WNWm6R5mi+hTlHRu3mNcX9EmBZYsU lS1ca4VYEU2TSkT92k817lQ8WauJShRJuG60vRjJl9ct3KNEIXe9UwAWIZVcPLSp18PC qDn'tuJctmu0Bdx87jEYVDn3j3MygRavcmr8a+VTPoywWfwdZmhtABpcEKg4l0k xvPg==

7. Analyze the 'Message-ID'

Message-ID	<VB2JSn31pzn_NFVs8yKaIA@notifications.google.com>
Subject	Security alert

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.

- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Headers Found	
Header Name	Header Value
Delivered-To	99240041213@klu.ac.in
X-Received	by 2002 a17-90b 2c88 b0-380 f85c 94b4 with SMTP id 98e67ed59e1d1-3966d20e7aamr18203644a91.7.1787759208712; Wed, 26 Aug 2026 08:46:48 -0700 (PDT)
ARC-Seal	i=1; a=rsa-sha256, t=1787759208, cv=none, d=google.com, s=arc-20260327, b=FzOkYeiqKQIBdpA6AzCyYACMzKubWbnlsjwsuVfQbwc7pXF44vBID79qGzKeXZrb/ CXlpTMyKCOR9GZmcOAwJh1zWM35OAYoA9IOESzW TgVUuGhH9DDiraCaEgRkKEXZZ 0QKyORXX3KnoOp7+n3gBVUYrmmN7JvVSqAWBDBZwcZeJ1E9vh63o8EqT99ZpoLUS 0uOX17DuFcuMnbleOyDxmVgZuP9pJC5t6PBHQv84JxysvHlV533GDu9NXC0wsa VnTV0 37LoAAK8b7Zoh3brXus9RIA1RjIMu770ToaSubdH11m4EHhHSF4iqcsVSH0 x9nw==
ARC-Message-Signature	i=1; a=rsa-sha256, c=relaxed/relaxed, d=google.com, s=arc-20260327, h=to:from:subject:message-id:gmail-feedback-id:date:mime-version:dkim-signature; bh=anf4m0bUILLhzrIS/6q9IKCjp5BFNfHmSF80wCkDY=; fh=kY H8ygDp5G665RpiHxWESRm8X4iWNczbAs51KfS=; b=Zot7Y1fghuHrCMp9khc/A+eS4+egjSjRW8R+oRaCLHh0oECIq5eVFvkb4WJwJW IOX8fMmMjUzF+m3DFI0yCuLEnu3EaiHgrbQlcj8UDGD0rum+RHQA5CPf8yL+4 N8R c3Z7YnKih9dLhFI0ubDLbZzJpDfYxHPZgOgVW4weE3uV8EKrig71X0DsN4pX+Pw5o6 4FmczVxKkyX0DBulsx3Hwusw2UVPPLXUF63PhTVRzviU74UjTU8WWhbOm+7YDDwpNmg ksFg43mwa8gm2rOp6vq5S8mEoPs K5sIRXW0JgVshAd8XBcmnhwyOHaiSiBe0QD GXaw==, dara=google.com
ARC-Authentication-Results	i=1; mx.google.com; dkim=pass header.i=@accounts.google.com header.s=20251104 header.b=GF9MztZa; spf=pass (google.com: domain of 3aapaggtlcxU89-cza6jvxx9f8ed 19916z x97uunplpmmmo56f vx.38@gaia.bounce.s.google.com designates 209.85.220.73 as permitted sender) smtp.mailfrom=3aAqPaggTCxU89-CzA6Jvxx9f8ED 19916z x97UUNPLLPMMNO56F vx.38@gaia.bounces.google.com; dmarc=pass (p=REJECT sp=REJECT di s=NONE) header.from=accounts.google.com; dara=neutral header.i=@klu.ac.in
Return-Path	<3aAqPaggTCxU89-CzA6Jvxx9f8ED 19916z x97UUNPLLPMMNO56F vx.38@gaia.bounces.google.com>
Received-SPF	pass (google.com: domain of 3aapaggtlcxU89-cza6jvxx9f8ed 19916z x97uunplpmmmo56f vx.38@gaia.bounces.google.com designates 209.85.220.73 as permitted sender) client-ip=209.85.220.73;
Authentication-Results	mx.google.com; dkim=pass header.i=@accounts.google.com header.s=20251104 header.b=GF9MztZa; spf=pass (google.com: domain of 3aapaggtlcxU89-cza6jvxx9f8ed 19916z x97uunplpmmmo56f vx.38@gaia.bounces.google.com designates 209.85.220.73 as permitted sender) smtp.mailfrom=3aAqPaggTCxU89-CzA6Jvxx9f8ED 19916z x97UUNPLLPMMNO56F vx.38@gaia.bounces.google.com; dmarc=pass (p=REJECT sp=REJECT dis=NONE) header.from=accounts.google.com; dara=neutral header.i=@klu.ac.in
DKIM-Signature	v=1; a=rsa-sha256, c=relaxed/relaxed, d=accounts.google.com, s=20251104, t=1787759208, x=1788364008, dm=klu.ac.in, h=content-type:to:from:subject:message-id:gmail-feedback-id:date:mime-version:from:to:cc:subject:date:message-id:reply-to:content-type; bh=anf4m0bUILLhzrIS/6q9IKCjp5BFNfHmSF80wCkDY=; b=GF9MztZaXlU+ZYnENcpfh6xgnyVb8DNikmwVfOTG5PeVxkZKzQ65f0syuLkqyRm1 38dZuSa+BSBCYs+6YW7oq WGT1b/wm5Iz960u+MeKIR+BSsEX8AluL8EZLSWneL7XR w8wC1/hHBQ19luELPBLWVl+ma47XvUAPvAOzVtobc3kczY9DmBcvZzm3Fic59IDHT JnCJXDcq7H69d7phgFkKZ77hvcCOP5v5VXI45a+qJpGy6hu+SwbBar 9A1QWoyzu ZKyxQEXsWmXDDsc6Qhccqy1lB01xfVjpuGU0FYs+lozbAupccDPzVnrOSqW6rQ HzUQ==
X-Google-DKIM-Signature	v=1; a=rsa-sha256, c=relaxed/relaxed, d=1e100.net, s=20251104, t=1787759208, x=1788364008, h=content-type:to:from:subject:message-id:gmail-feedback-id:date:mime-version:x-gm-message-state:from:to:cc:subject:date:message-id:reply-to:content-type; bh=anf4m0bUILLhzrIS/6q9IKCjp5BFNfHmSF80wCkDY=; b=CL2e6ymtePGRGmsMubOXctdbE2Aq5/L49zInof52eA4MTvTHNX1O5t5d60xz19r6 R04bskDnvd4IY98yIH+keoy/crJ1nsApq Cw3wcQubhP5FNWKCWyzc39GuQAkJW+DU YMAUNgNz6lqxQ2wu5HNzQ34waBdlQ8CnMc5WNWim6R5mi+hTHRU3mNcX9EmBZYsU IS1ca4VYEU2TSK792k817Q8WaujShRJUG60vRjll9c3KNEDE9JwaAWZVCPLsp 18PC qDn1uJcfcuOtBte87JEYVn3j3MygRavcmr8a+VTpoywIWFwdZmhITABpcEKg4l0lk xvPg==
MIME-Version	1.0
Date	Wed, 26 Aug 2026 15:46:45 GMT
X-Account-Notification-Type	325
Feedback-ID	325 account-notifier
gmail	true
X-Google-Notification-Metadata	CgiSAA
X-Notifications	cbdea11f749e0000
X-Notifications-Bounce-Info	AWKMoMQL1cnQJGSYYstRv-tMBnAh2o7VQ9g8i5Smy4AB3_-AJ8koagQmWfXpJF2nN9nYDRBSSYCYG72YQOqghdJTLTYUN_XvBdZDSxj0D0R1oeUtnIgo3LlIT7T2H9Zq90kujN1zq13v7eWQDDJ2VP4-WKNaIPXcLWWpyfn x6zvpVPlwW8j-IR9G7WYpUpZe8haXFsvYhBjWIGNAKJHlOhACj26j4W46nh0D_IZNKIm-NusxQA8ob2pwZwJdK5pQJyqPKUNjAwNjAOMDQxNTM1NTK2OTMzmIg
Message-ID	<VB2Jsn31pzn_NFVs8yKaIA@notifications.google.com>
Subject	Security alert
From	Google <no-reply@accounts.google.com>
To	99240041213@klu.ac.in
Content-Type	multipart/alternative; boundary="0000000000008c470e0659f52021"

